

Ethics of New Technologies

When Algorithms Discriminate:
The PATROL-AI Case Study

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Two Frameworks, One Goal

We selected the ACM and IEEE codes - the two most widely adopted ethical frameworks in computing and engineering.

ACM Code of Ethics

Association for Computing Machinery · 2018

- **Computing-specific**
- Detailed sub-principles organized by tier
- Covers algorithmic fairness & privacy
- Obligations for both individuals & organizations

IEEE Code of Ethics

Institute of Electrical & Electronics Engineers · 2020

- **All engineering disciplines**
- 10 concise, universal clauses
- Public safety as paramount principle
- Emphasizes honesty and fair treatment

The 5 Principles Relevant to Our Case

1.1

Contribute to well-being

All stakeholders must benefit - not just the majority.

1.2

Avoid harm

Identify and mitigate negative consequences, even unintended.

1.4

Be fair - no discrimination

Explicitly addresses algorithmic bias and unfair outcomes.

1.6

Respect privacy

Consent is required before using personal data.

2.5

Thorough evaluation

Anticipate and report unintended consequences of systems.

IEEE Clauses & Side-by-Side Comparison

1

Clause 1

Public safety and welfare are paramount above all other considerations.

7

Clause 7

Accept honest criticism and acknowledge errors when discovered.

8

Clause 8

Treat all persons fairly regardless of origin, ethnicity, or social background.

Key Differences at a Glance

Feature	ACM (2018)	IEEE (2020)
Scope	Computing only	All engineering
Structure	Extensive sub-principles	10 concise clauses
Algorithmic Bias	Explicit (Principle 1.4)	Implicit (Clause 8)
Privacy	Dedicated principle (1.6)	Implied

CASE STUDY

PATROL-AI

Predictive Policing in a City of 500,000

-18%

Crime rate drop

Across the city

-12%

Response time

Emergency services

€900k

Annual savings

City budget

A fictional-but-realistic scenario: a tech startup deploys an AI algorithm to predict and prevent crime. The headline numbers look impressive - but the story doesn't end there.

Marta's Audit Reveals Structural Bias

3.4x

Higher flagging rate
in 3 neighbourhoods
(15,000 residents)

Historical Bias in Training Data

Past over-policing of low-income & immigrant neighbourhoods fed directly into the model's predictions.

Compounding Feedback Loop

More flags → more patrols → more arrests → more data points → even more flags. The bias amplifies itself.

GDPR Violation

CCTV footage was used without proper consent - a clear breach of Principle 1.6 (ACM) and data protection law.

Management Dismisses the Audit

Marta reports her findings. Response from management: "The numbers speak for themselves."

Verdict: ACM & IEEE Applied to PATROL-AI

ACM 1.1 VIOLATED**Contribute to well-being**

15,000 stakeholders are ignored - their welfare is sacrificed.

ACM 1.2 VIOLATED**Avoid harm**

Documented harm is deliberately continued by management.

ACM 1.4 VIOLATED**Non-discrimination**

Algorithmic discrimination by race and income is clear.

ACM 1.6 VIOLATED**Respect privacy**

CCTV data used without GDPR-compliant consent.

ACM 2.5 UPHELD**Thorough evaluation**

Marta's audit was thorough - management who failed this, not Marta.

IEEE 1 VIOLATED**Public welfare paramount**

Minority welfare is sacrificed for aggregate statistics.

IEEE 7 VIOLATED**Accept honest criticism**

Management rejected documented, timestamped findings.

IEEE 8 VIOLATED**Fair treatment**

Discriminatory outcomes against a minority community.

Does "Greatest Good" Justify the Harm?

Management's argument:

"An 18% crime reduction for 485,000 people outweighs complaints from 15,000."

1

Long-term calculus (Mill)

The feedback loop means harm compounds over time. 5-year cumulative harm to 15,000 likely exceeds marginal benefit to 485,000.

2

Rule utilitarianism

The rule 'sacrifice minority civil liberties for statistics' is catastrophic if adopted universally. Good rules must be safe to generalize.

3

Intensity of harm (Bentham)

Severe concentrated harm (harassment, loss of freedom) is not offset by diffuse marginal benefit spread across a much larger population.

What Should Marta Do?

Both codes and utilitarian philosophy converge on the same answer: she must act.

01 Formal Documentation

Record all findings with timestamps, data, and evidence. A verbal report is not enough - create a written audit trail.

02 Propose Technical Fixes

Bias correction algorithms, fairness constraints, and independent auditing are all feasible solutions. Present them alongside the problem.

03 Internal Escalation

Raise the issue beyond immediate management - to the ethics board, legal team, or senior leadership. Document every response.

04 External Reporting (if needed)

If internal channels fail, contact the national data protection authority. GDPR gives regulators the power to compel action.

Key Takeaways

1

PATROL-AI as deployed is ethically unacceptable - not because it has no benefits, but because it achieves them by concentrating harm on a specific community.

2

ACM, IEEE, and utilitarian philosophy all converge on the same verdict - seven clauses violated, one upheld.

3

Aggregate benefit does not justify unjust distribution of harm. Ethics must ask not only how much good is produced, but who bears the cost.

4

The ethical engineer's responsibility does not end at deployment. Ongoing auditing, transparency, and willingness to escalate are as important as the initial design.